

JAHNAVI VASALA

DATA ENGINEER

+1 (505) 316-8548 | SAN JOSE, CA | [| jaanviv11@gmail.com](mailto:jaanviv11@gmail.com)

PROFESSIONAL SUMMARY

Data Engineer with 5 years of progressive experience designing and optimizing scalable batch and streaming data platforms across consulting, enterprise software, and fintech environments. Strong expertise in building ETL/ELT pipelines using Spark, Airflow, and SQL while implementing dimensional warehouse architectures in AWS and Azure ecosystems. Experienced in developing incremental ingestion and CDC-based workflows to efficiently manage high-volume structured and semi-structured datasets. Proficient in performance tuning, distributed processing optimization, and warehouse cost management strategies. Skilled in implementing schema governance, version-controlled data contracts, and validation frameworks to ensure production reliability and data integrity. Adept at building analytics-ready and feature-ready datasets that support reporting, operational intelligence, and predictive modeling initiatives. Proven collaborator with engineering, analytics, and product teams to deliver secure, resilient, and business-aligned data solutions.

SKILLS

- **Programming and Query Languages:** Python, SQL, PySpark, Scala, Shell scripting
- **Big Data and Distributed Processing:** Apache Spark, Spark SQL, Databricks, Hadoop, MapReduce
- **Streaming and Messaging:** Apache Kafka, Spark Structured Streaming, Event-driven architectures
- **Cloud Platforms :** AWS (S3, EMR, Glue, Redshift, Lambda, Athena, IAM, CloudWatch) Azure (Data Factory, Synapse Analytics, ADLS Gen2, Databricks, Azure Monitor, Key Vault)
- **Data Warehousing and Storage :** Snowflake, Amazon Redshift, Azure Synapse, Data Lakes, Parquet, ORC
- **Workflow Orchestration :** Apache Airflow, Azure Data Factory, Cron scheduling
- **Data Modeling and Architecture:** Dimensional modeling, Star schema, Snowflake schema, Data normalization, Data lake architecture, ETL and ELT frameworks
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Data Quality and Governance:** Great Expectations, Data validation frameworks, Data lineage documentation, Access control, Encryption standards
- **DevOps and Infrastructure:** Git, Jenkins, CI/CD pipelines, Docker, Terraform
- **Analytics and BI Integration:** Power BI, Tableau, ML feature engineering support

EXPERIENCE

INTUIT – Data Engineer
San Diego, CA

(Aug 2025- present)

- Designed distributed Spark-based ETL pipelines on AWS EMR processing high-volume financial transaction datasets
- Architected and maintained 40+ Airflow DAGs orchestrating ingestion from APIs and S3 into Snowflake
- Implemented CDC-based incremental ingestion strategies eliminating full-table refresh dependencies
- Optimized Snowflake clustering and query plans improving reporting performance by 34%
- Built Kafka-based streaming pipelines supporting near real-time financial event processing
- Developed transformation models using dbt with version-controlled testing to ensure data integrity across warehouse layers
- Designed dimensional data marts for revenue, tax, and customer analytics
- Implemented reconciliation and validation frameworks reducing financial data discrepancies by 36%

- Conducted root cause analysis and implemented structured backfill processes for production failures
- Tuned Spark executor configurations and partition strategies to improve transformation workloads
- Implemented encryption and IAM-based RBAC controls for secure handling of sensitive financial data
- Automated infrastructure provisioning with Terraform reducing environment setup time to under 2 hours
- Improved monitoring and alerting mechanisms maintaining 99.8% SLA adherence
- Collaborated with backend teams to define version-controlled data contracts and manage schema evolution

SAP – Data Engineer
Plano, Texas

(Jan 2023-Aug 2025)

- Designed enterprise-scale ETL pipelines integrating ERP and finance datasets into AWS Redshift
- Developed dimensional data models supporting enterprise analytics and reporting use cases
- Optimized SQL queries and indexing strategies improving dashboard performance by 31%
- Implemented incremental ingestion strategies reducing batch processing time by 33%
- Built ingestion workflows using Azure Data Factory to integrate ERP datasets into Azure Data Lake Storage
- Developed transformation pipelines within Azure Synapse supporting procurement and finance analytics
- Managed structured staging and curated storage layers across AWS S3 and Azure ADLS
- Designed validation and reconciliation workflows reducing reporting discrepancies by 28%
- Improved enterprise reporting latency from 24 hours to under 8 hours
- Implemented IAM and RBAC-based access controls across AWS and Azure environments
- Developed monitoring dashboards maintaining 99.5% pipeline reliability
- Assisted migration from legacy on-prem systems to hybrid cloud warehouse architecture
- Defined and enforced schema evolution policies using version-controlled data contracts between upstream ERP systems and warehouse ingestion pipelines
- Designed controlled backfill strategies to reload historical datasets without disrupting downstream consumers
- Implemented unit testing and data validation checks for transformation logic prior to warehouse loads

CAPGEMINI– Data Engineer
Bangalore, India

(March 2019- Nov2021)

- Assisted in developing ETL pipelines ingesting structured client datasets into centralized warehouse environments
- Built SQL transformation logic and aggregation queries supporting client reporting solutions
- Supported migration of on-prem datasets to AWS S3 and Redshift
- Implemented data cleansing and validation scripts reducing manual reporting errors by 22%
- Developed Python automation scripts to streamline recurring data processing workflows
- Participated in dimensional model design for reporting and analytics systems
- Assisted in debugging and resolving failed pipeline executions
- Improved batch processing efficiency by 18% through SQL optimization
- Collaborated with client stakeholders to gather reporting and transformation requirements
- Documented data flow architecture and transformation logic for knowledge transfer
- Supported implementation of role-based data access controls
- Participated in production issue resolution and system stabilization efforts

EDUCATION

Master of Science in Computer Science | Cleveland State University, OH.
BTech in Electrical and Electronics | NIT GOA,INDIA